Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | Industrial Inspection, IoT & Edge Deployment

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Professional Summary

Computer-vision engineer with 5.5 years leading production CV/IoT systems at MAS Holdings, South Asia's largest apparel manufacturer, including a COVID-19 remote-vitals CV device deployed across hospital wards and recognized by the Government Medical Officers' Association (GMOA) of Sri Lanka – his one direct clinical CV deployment. Complements this with a PhD (University of Trento) built on rigorous benchmark and ablation design, quantifying accuracy/latency/CPU trade-offs under hard real-time constraints, plus human-centred evaluation work directly relevant to validating a CV system for clinical use. Canadian Permanent Resident based in Toronto; open to relocating to Burnaby/Vancouver.

Core Competencies

Computer Vision & Perception: OpenCV (template and feature matching, image registration and warping, corner/edge detection, explainable defect overlays); industrial and microscopic camera-array inspection; real-time embedded CV pipelines; sensor/IoT fusion.
Evaluation & Validation Rigor: frozen-protocol benchmark suite design, ablation studies, statistical evaluation of accuracy/latency/CPU trade-offs, human-centred usability evaluation.

Technical Stack

C++17, Python (TensorFlow, Keras, PyTorch, scikit-learn), OpenCV, embedded/edge deployment (Raspberry Pi, ARM CPU profiling), real-time inference under hard latency budgets, AWS, SQL, ETL pipelines, Docker, CI/CD, Git.

Education

PhD, Information Engineering and Computer Science – University of Trento, Italy 2021 - 2025
MSc, Telecommunication and Electronic Engineering – Sheffield Hallam University, UK 2016 - 2018
BEng (Hons), Electronic Engineering – Sheffield Hallam University, UK 2013 - 2014

Research and Work Experience

Independent Researcher | *Self-directed, Toronto, Canada* *Jan 2026 - Present*

- Two peer-reviewed papers accepted (Smart Drums, JAES; MIRaaS, IEEE IS2).
- Maintains Nebula, a C++17 audio-feature-extraction library with per-feature ARM/Raspberry Pi latency benchmarks.

Postdoctoral Researcher | *University of Trento, Trento, Italy* *Jan 2025 - Dec 2025*

- Owned the full ML pipeline for streaming audio + IMU/gesture data: acquisition, DSP/feature engineering, training/eval, edge deployment (Raspberry Pi 4), monitoring.
- Published real-time detector: 14 ms inference on Raspberry Pi 4, F1 0.76, 31.4% CPU, 74x faster than a DTW baseline (JAES 2026).
- Designed frozen-protocol benchmark suites and ablations quantifying accuracy/latency/CPU trade-offs – an evaluation methodology directly transferable to validating a medical CV system before clinical deployment.

Visiting Researcher | *McGill University (IDMIL/CIRMMT), Montreal, Canada* *May 2024 - Aug 2024*

- Built a self-powered smart electric guitar (BNO055 IMU + HiFiBerry + Raspberry Pi 4). Fused IMU and audio signals via a hierarchical FSM, F1 0.78 on a 500-event corpus (IEEE IS2 2025).

Visiting Researcher | *University of Visual and Performing Arts, Colombo, Sri Lanka* *Jan 2024 - Mar 2024*

- Ran a human-centred evaluation of musicians' perception of smart instrument pattern detection (IEEE I3DA 2025), combining quantitative and qualitative methods – directly relevant to validating a clinical tool's real-world usability.

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka* *Aug 2020 - Feb 2021*

- Built ML and statistical pipelines for financial forensics, compliance monitoring, and anomaly detection on large structured enterprise data.

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka* *Jan 2015 - Jul 2020*

- Owned end-to-end computer-vision and IoT systems for manufacturing at scale, South Asia's largest apparel manufacturer, taking camera/sensor/IoT streams into production workflows; operational efficiency gains up to +300%.
- Built and deployed a COVID-19 remote-vitals computer-vision device across hospital wards, digitizing patient vitals under real clinical adoption stakes; deployment recognized by the Government Medical Officers' Association (GMOA) of Sri Lanka.
- Automated multilingual care-label QC using OpenCV template and feature matching with an explainable defect overlay, integrated with conveyor and PLC reject hardware; cut inspection time by 99.5%.
- Built loom-side fabric structural-defect detection using a microscopic camera array; cut operator headcount by 50%.
- Ran real-time IoT ETL across sites in South Asia, North America, and Africa, with C++/Python inference on embedded hardware.
- Introduced CI/CD and code review practices; mentored engineers across the team.

Publications

Silva, N. & Turchet, L. (2026). Real-Time Audio Pattern Detection for Smart Musical Instruments. *Journal of the Audio Engineering Society*, 74(3). Silva, N. & Turchet, L. (2026, in press). Smart Drums. *Journal of the Audio Engineering Society*. Silva, N. & Turchet, L. (2022). An SSIM-Based Training-Free Detection Method. *Proceedings of DAFx*, achieving 95% detection accuracy without training data.

Highlights

GMOA (Government Medical Officers' Association, Sri Lanka) recognition, 2020, for a COVID-19 remote-vitals computer-vision device deployed across hospital wards during the pandemic. MIDI Innovation Awards 2023 Finalist ("Hot Licks", Prototypes & Non-Commercial Software).